1. **Find the first non-repeating character in a string using HashMap**
   Code:

```
package Day8;

import java.util.*;

public class FirstNonRepeatingChar {

    public static Character findFirstNonRepeating(String str) {
        Map<Character, Integer> freqMap = new LinkedHashMap<>();

        for (char ch : str.toCharArray()) {
            freqMap.put(ch, freqMap.getOrDefault(ch, 0) + 1);
        }

        for (Map.Entry<Character, Integer> entry : freqMap.entrySet()) {
            if (entry.getValue() == 1) {
                return entry.getKey();
            }
        }
        return null; // No non-repeating character found
    }

    public static void main(String[] args) {
        String input = "swiss";
        Character result = findFirstNonRepeating(input);
        System.out.println("First Non-Repeating Character: " + result);
    }
}
```

2. **Use List for cart, Map for product-price mapping.**
   Code:

```
package Day8;

import java.util.*;

public class OnlineShoppingCart {

    public static void main(String[] args) {
        List<String> cart = new ArrayList<>();
        Map<String, Double> productPrice = new HashMap<>();

        // Product catalog
        productPrice.put("Laptop", 75000.0);
        productPrice.put("Phone", 30000.0);
        productPrice.put("Headphones", 2000.0);

        // Add items to cart
        cart.add("Laptop");
        cart.add("Headphones");
        cart.add("Phone");

        // Calculate total
```

```
        double total = 0;
        for (String item : cart) {
            total += productPrice.get(item);
        }

        System.out.println("Cart Items: " + cart);
        System.out.println("Total Price: ₹" + total);
    }
}
```

3. **Use Map<Book, Boolean> to track availability.**
   Code:

```
package Day8;

import java.util.*;

class Book {
    String title;
    String author;

    Book(String title, String author) {
        this.title = title;
        this.author = author;
    }

    @Override
    public String toString() {
        return title + " by " + author;
    }

    @Override
    public boolean equals(Object obj) {
        if (this == obj) return true;
        if (!(obj instanceof Book)) return false;
        Book other = (Book) obj;
        return Objects.equals(title, other.title) && Objects.equals(author, other.author);
    }

    @Override
    public int hashCode() {
        return Objects.hash(title, author);
    }
}

public class LibraryManagement {

    public static void main(String[] args) {
        Map<Book, Boolean> library = new HashMap<>();

        Book b1 = new Book("1984", "George Orwell");
        Book b2 = new Book("The Alchemist", "Paulo Coelho");
        Book b3 = new Book("Clean Code", "Robert C. Martin");
```

```
        // Track availability
        library.put(b1, true);
        library.put(b2, false);
        library.put(b3, true);

        // Check availability
        for (Map.Entry<Book, Boolean> entry : library.entrySet()) {
            System.out.println(entry.getKey() + " - Available: " + entry.getValue());
        }
    }
}
```